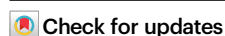


Machine-learning-guided tungsten single atoms promote oxyhydroxides for noble-metal-free water electrolysis

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Lowering the overpotential of oxygen evolution reaction with electrocatalysts is essential for efficient renewable-electricity-driven electrolysis. Active noble-metal catalysts suffer from leaching and scarcity, while non-noble alternatives face limited intrinsic activity. Here we combine computational guidance with experimental validation to identify atomically dispersed tungsten within NiFe oxyhydroxide, namely W_1 -NiFeOOH, as a promising noble-metal-free oxygen evolution reaction catalyst. An equivariant transformer-based machine-learning interatomic potential accelerates out-of-domain adsorption energy predictions and nominates W_1 -NiFeOOH from 3,976 single-atom-incorporated metal oxyhydroxide configurations. Cyclic-electrodeposited W_1 -NiFeOOH achieves a high current density of 13.1 A cm^{-2} at 2.0 V and remains stable for 500 hours in alkaline exchange-membrane water electrolysis with commercial membranes. In situ spectroscopy and density functional theory calculations suggest that subsurface W promoter induces synergistic electron redistribution at neighboring Ni-O-Fe edge active sites, thereby lowering the proton-coupled electron-transfer barrier for the deprotonation step and facilitating transformation into the active γ -phase. This integrated computational-experimental workflow provides a blueprint for cost-effective catalyst design for sustainable energy systems.

Oxygen evolution reaction (OER) is a fundamental oxidation process in renewable energy conversion, including electrocatalytic water splitting and carbon dioxide reduction^{1–3}. The overpotential is an excess voltage required beyond the

thermodynamic value for the reaction products, which acts as a kinetic barrier in the four-electron transfer process and reduces total energy conversion efficiency^{4,5}. Developing optimal electrocatalysts to minimize the overpotential is essential for

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lowering electricity consumption and ensuring the economic viability of renewable energy systems.

Iridium-based noble metal oxides are widely regarded as benchmark catalysts for OER due to their catalytic activity⁶. However, their scarcity and high expense limit large-scale applications, driving extensive research into cost-effective alternatives^{7,8}. Discovering non-noble metal catalysts might cost-effectively replace the demands on the precious metal, which require hazardous environmental impact during the mining process.

Among earth-abundant compounds, metal (oxy)hydroxides consisting of 3d transition metals, particularly Ni and Co, have shown considerable promise in alkaline OER environments^{9,10}. Multi-metallic (oxy)hydroxides, formed by incorporating additional elements, can tailor the electronic structure and the adsorption energies of reaction intermediates, enhancing catalytic performance^{11,12}. For instance, doping Fe into Ni (oxy)hydroxides yields reactive Ni–Fe layered double hydroxides^{13,14}. Introducing other 3d transition metals^{11,15,16}, high-valent 4d and 5d transition metals^{17–19} or nonmetal elements²⁰ has also effectively tailored the catalytic performance.

Recently, heterogeneous single-atom catalysts (SACs) have revolutionized catalyst design by featuring atomically dispersed metal atoms on supports, maximizing metal utilization and unlocking unconventional electronic structures^{21,22}. The absence of metal-metal bonds significantly influences the electrochemical microenvironment through strong interactions between single atoms and support materials^{23,24}. Integrating SACs with metal (oxy)hydroxides creates a synergistic next-generation catalyst platform with the potential to surpass conventional catalytic performance limits.

Compared to the experimental approach, density function theory (DFT) calculations have been regarded as efficient for assessing candidates by acquiring fundamental material properties through electronic structure calculations²⁵. However, the extensive combinatorial space and structural complexity of SACs integrated with metal (oxy)hydroxides render conventional DFT-based approaches impractical due to high computational costs. To address these limitations, machine learning (ML) techniques offer a promising strategy by predicting material properties directly from atomic configurations, thus circumventing computationally intensive processes^{26,27}. Among advanced ML models, EquiformerV2²⁸, a state-of-the-art graph neural network with an equivariant transformer model, stands out for maintaining rotational and translational invariance, enabling accurate and rapid predictions. Accompanied by recent advances in graphics processing unit (GPU) technology for parallel computing, ML models reduce the time and computational resources requirement, allowing efficient high-throughput screening of complex catalyst candidates^{29–33}.

In this study, we employed a ML-guided screening strategy using the pretrained EquiformerV2 model to identify NiFe oxyhydroxides with atomically dispersed W atoms (W₁–NiFeOOH) as the most promising noble metal-free OER catalyst candidates. The ML-guided catalyst candidates were experimentally prepared via scalable cyclic electrodeposition, resulting in a substitutional octahedral W atomic local structure. Electrochemical assessments showed that noble metal-free W₁–NiFeOOH exhibited significant OER performance, delivering a high current density of 13.1 A cm^{−2} at 2.0 V (80 °C) in AEMWE employing commercial membranes, thereby setting a benchmark for noble metal-free anode catalysts under practical water electrolysis conditions. In situ spectroscopies and computational calculations provide insights into the distinct catalytic mechanisms and active motifs. Anchored Ir atoms directly act as active sites, while substituted W atoms function as modulators, promoting the original active sites in the NiFeOOH edge plane. This modulation stabilizes the deprotonation at the active site to lower the energy barrier and promotes the structural transformation to the active γ -phase at lower potentials. These combinations of computational predictions and experimental

validation accelerate the discovery of competitive noble-metal-free catalytic solutions towards sustainable energy conversion with the renewable electricity.

Results

Machine learning-guided catalyst screening

To accelerate the discovery of efficient noble-metal-free OER catalysts, we employed a ML framework, EquiformerV2²⁸, a large-parameter equivariant graph neural network pretrained on the OC22 dataset. This ML-guided approach predicts total energies directly from atomic structures without solving computationally intensive Kohn–Sham equations, enabling rapid evaluation across large materials candidate spaces (Fig. 1a).

We constructed single-atom incorporated metal oxyhydroxide slab models (Supplementary Data 1) within an established K⁺-intercalated γ -phase oxyhydroxide matrix³⁴, creating a combinatorial candidate space from 8 different metal oxyhydroxides (M_AM_BOOH) with 41 elements of single-atom (M₁) and 13 types of possible adsorption sites on the edge (O1–12) and basal (0001) plane, resulting in 3976 configurational candidate models (M₁–M_AM_BOOH) (Supplementary Fig. 1). OER intermediates (*V, *OH, *O, and *OOH) were considered for each configuration, resulting in over 15,000 adsorbed structures and these geometries represent catalytically relevant systems not included in the original OC22 training domain. To validate model performance on this out-of-domain system class, we generated a custom DFT dataset by selecting structures corresponding to four representative active motifs illustrated in Fig. 1b, derived from unary NiOOH and binary NiFeOOH systems, which were subsequently used for DFT validation and mechanistic investigation (Supplementary Fig. 2).

We benchmarked several state-of-the-art GNN models, including GemNet, PaiNN, and DimeNet++ on our custom DFT-relaxed dataset. By leveraging rapid GPU-accelerated predictions, EquiformerV2 exhibited the highest accuracy in adsorption energy prediction while achieving an approximately three-order-of-magnitude faster adsorption energy evaluation than conventional DFT calculations (Supplementary Fig. 3). Based on these results, EquiformerV2 was selected as the backbone model for further screening. To further verify the model fidelity and enable domain-specific transferability, we fine-tuned the backbone architecture using a curated subset of the M₁–NiNiOOH dataset. This approach significantly reduced total energy prediction error MAE less than 0.01 eV atom^{−1} (Supplementary Figs. 5 and 6) and enhanced accuracy in estimating adsorption energy differences (Supplementary Fig. 7). Notably, the fine-tuned model achieved adsorption energy MAEs below 0.2 eV for unseen binary M₁–NiFeOOH systems (Supplementary Fig. 8), validating its transferability across compositional domains.

High-throughput screening of all M₁–M_AM_BOOH candidates was performed using the original pretrained EquiformerV2 backbone model, which rapidly evaluated total energies and atomic forces of OER intermediate-adsorbed structures via iterative relaxation. Adsorption energy distributions across the full M_AM_BOOH matrix were visualized using violin plots (Supplementary Fig. 9), revealing that NiFeOOH and CoFeOOH-based systems consistently exhibited the lowest average energy barriers for the potential-determining step, in agreement with trends reported in previous experimental studies. A linear scaling relation for predicted *OOH and *OH adsorption energies was rediscovered, closely aligning with previous DFT-derived results and varying with the adsorption site configuration (Supplementary Fig. 10)^{4,35}. The two-dimensional activity plot illustrated the adsorption energy landscape for the candidates, and the contour plots were constructed based on the scaling relations from the substitutional adsorption configurations (Fig. 1c) and other configurations (Supplementary Figs. 11 and 12). The OER volcano plot was employed to definitively identify W₁–NiFeOOH and Ir₁–NiFeOOH as the most promising candidates with the lowest predicted overpotentials, surpassing the

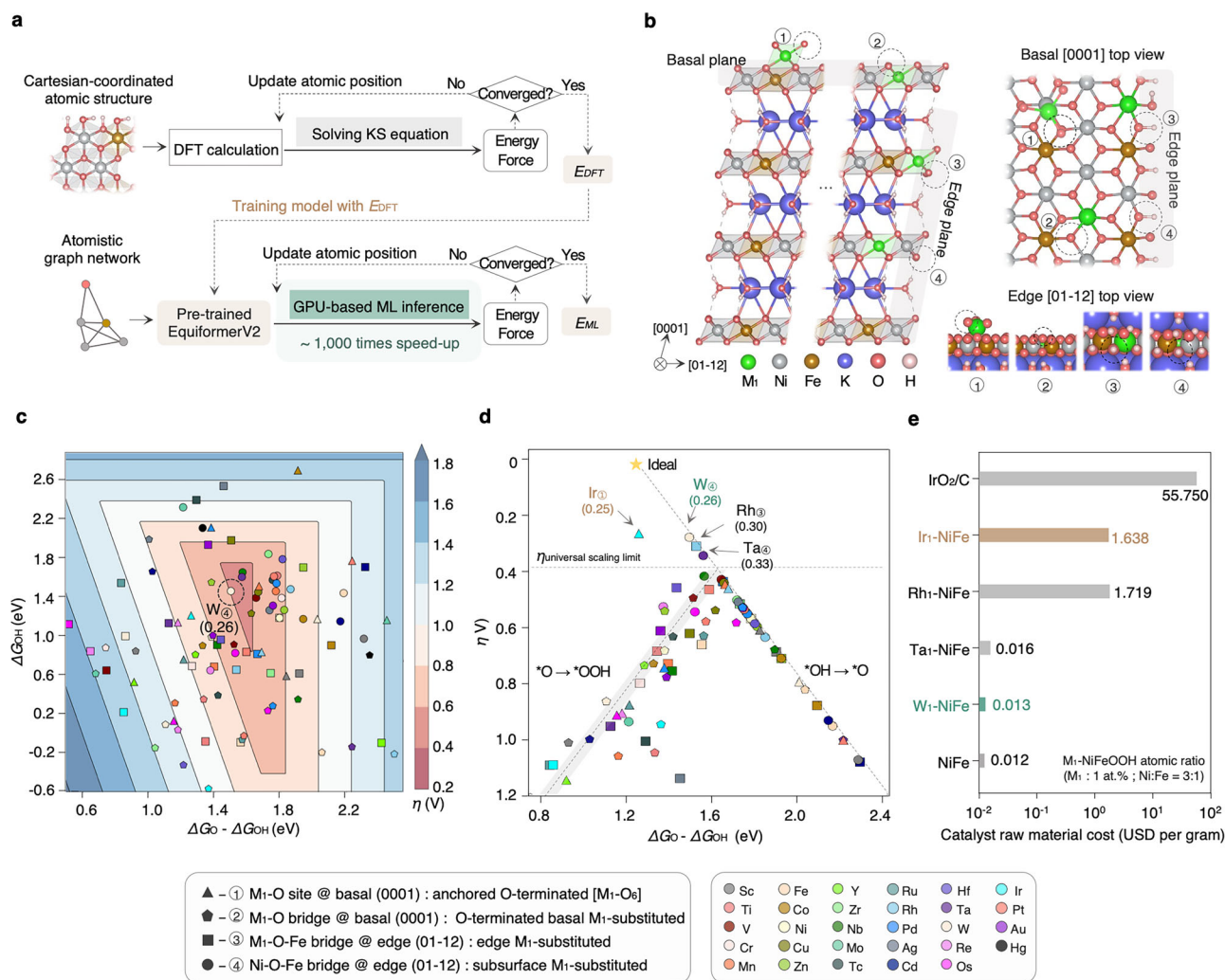


Fig. 1 | Machine learning-guided catalyst screening. **a** Schematic illustration of the catalyst screening process assisted with pre-trained graph neural network model, EquiformerV2. **b** Structural coordination models of four representative active motifs within the M_1 -NiFeOOH system. **c** OER activity volcano landscape shown as a 2D heat map of predicted overpotential (η) based on mapping to adsorption energies for OH^* (ΔG_{OH}) and O^* (ΔG_O). Each symbol represents for

different active site configuration. The linear scaling relationship of OOH^* from substitutional adsorption configurations was used for the contour plot. **d** 1D volcano plot showing predicted OER overpotential for W_1 -NiFeOOH and Ir_1 -NiFeOOH as promising candidates. **e** Comparison of raw materials costs per gram for various catalyst candidates based on the metallic element contents. Source data for Fig. 1c–e are provided as a Source data file.

universal theoretical limit (Fig. 1d). On the other hand, anchored Ir motifs on the basal (0001) plane were predicted to function as catalytic sites themselves, with the energy barrier on the $*O$ to $*OOH$ transition as the limiting step. The raw material cost analysis revealed that including noble metals in catalyst design, even at low concentrations as 1 at.%, significantly escalates the overall cost compared to non-precious metal alternatives, highlighting the importance of cost-effective non-noble catalysts (Fig. 1e).

Morphology and structural characterization

To experimentally validate our proposed candidates, we prepared W_1 -NiFeOOH and other variants using an optimized cyclic voltammetry (CV) electrodeposition technique, enabling scalable synthesis within minutes at 25 °C directly on electrode substrates (Supplementary Fig. 13). By scanning between -0.1 and -1.8 V vs. Ag/AgCl at 50 mV s⁻¹, the simultaneous electroreduction of homogeneously dissolved metal cations, including precursors for (oxy)hydroxides and low-concentration single-atom precursors, facilitates the rapid growth of metal (oxy)hydroxides with atomically dispersed W atoms while suppressing the formation of undesired metal oxides and nanoparticles. The resulting edge-rich nanosheet morphology was

confirmed through field-emission scanning electron microscopy and compared with the sample prepared via constant-potential electrodeposition (Supplementary Fig. 14). In addition, the optimized CV electrodeposition approach efficiently removes excess attached metal ions during the CV sweeps, may resulting in a more homogeneous atomically dispersed phase. High-angle annular dark field-scanning transmission electron microscopy (HAADF-STEM) images revealed atomically dispersed W atoms within NiFe oxyhydroxide nanosheet, exhibiting significantly greater brightness than neighboring Ni and Fe atoms due to stronger scattering from higher atomic number elements³⁶ (Fig. 2a and Supplementary Fig. 15). The atomic distribution of W species was further confirmed by energy-dispersive X-ray spectroscopy (EDX) elemental mapping (Fig. 2b) and extended intensity line profiles (Supplementary Fig. 16). Compositional evaluation using inductively coupled plasma atomic emission spectrometry (ICP-AES) determined that W accounted for an atomic ratio of 2.6% of the total metal content (Supplementary Table 2). Atomic force microscopy (AFM) images revealed that a single layer of W_1 -NiFeOOH nanosheet had a thickness of approximately 4 nm (Fig. 2c). The stacked structure of these thin layers enhances the exposure of edge active sites (Supplementary Fig. 17). We prepared

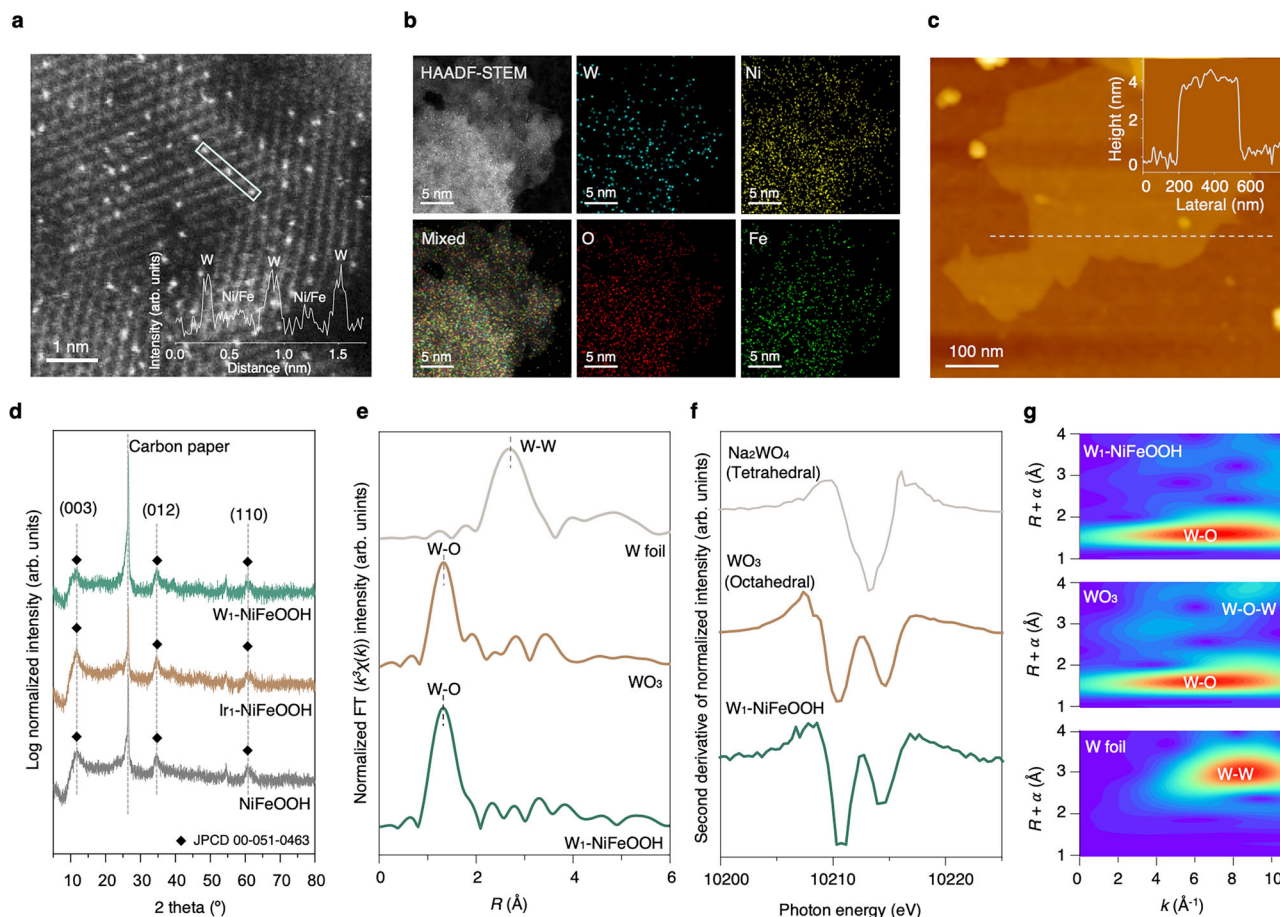


Fig. 2 | Morphology and structural characterization. **a** HAADF-STEM image showing atomically dispersed W atoms in NiFe oxyhydroxide lattice. W atoms with higher atomic number were brighter than Ni or Fe atoms, verified with the line intensity profile (inset). **b** EDX mapping representing the elemental distribution of W, Ni, Fe, and O. **c** AFM image showing a single W_1 -NiFeOOH nanosheet layer with the height profile (inset). **d** XRD patterns of M_1 -NiFeOOH samples on carbon paper.

Diamond symbols denote the reflections from the NiFe (oxy)hydroxide phase. **e** k^3 -weighted $W L_3$ -edge FT-EXAFS spectra in R -space, where R denotes the phase-unshifted distance. **f** Second derivatives of normalized $W L_3$ -edge XANES spectra. **g** WT analysis of k^3 -weighted EXAFS for W_1 -NiFeOOH and reference samples, where k is the photoelectron wavevector, and $R + \alpha$ is the phase-shifted distance. Source data for Fig. 2a, c–f are provided as a Source data file.

other variants and confirmed the incorporation of atomically dispersed atoms within the metal (oxy)hydroxides, demonstrating the universality of the electrodeposition method (Supplementary Figs. 18–20). X-ray diffraction (XRD) patterns indicated that the crystalline structure of electrodeposited pristine (oxy)hydroxides was retained with the incorporation of single atoms (Fig. 2d and Supplementary Fig. 21), with diffraction peaks consistent with the crystalline domains observed in TEM analysis. The phase consistency of the (oxy)hydroxides was further evidenced by characteristic band peaks in Raman spectra (Supplementary Fig. 22)^{37,38}.

In addition, X-ray photoelectron spectroscopy (XPS) was performed to examine the surface electronic structure and chemical configuration. The $W 4f$ core-level spectra showed distinct peaks corresponding to $W 4f_{7/2}$, which could be deconvoluted into two sets of doublets attributed to W^{5+} and W^{6+} with electron transfer to the oxyhydroxide host (Supplementary Fig. 23)³⁹. The local atomic environment and electronic structure were further investigated by X-ray absorption spectroscopy (XAS). In the case of X-ray absorption near edge structure (XANES) at the L_3 -edge of $5d$ transition metals, the whiteline energy is highly sensitive to the valence state and empty d orbitals states⁴⁰. The $W L_3$ -edge XANES spectra of W_1 -NiFeOOH showed that the whiteline position was higher than W foil and closely aligned with WO_3 (Supplementary Fig. 24), indicating W oxidation state close to +6. The higher whiteline intensity compared to WO_3

demonstrated a distorted WO_6 octahedral symmetry due to local substitutional lattice distortion⁴¹.

The Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) spectra of W_1 -NiFeOOH exhibited a distinct peak for the $W-O$ scattering path in the first shell (Fig. 2e). The absence of a peak corresponding to the first-shell $W-W$ path observed in the W foil indicated no detectable $W-W$ bonds and confirmed the atomically dispersed state of W atoms. The weak second-shell scattering peaks could be attributed to $W-Ni/Fe$ bonds of the distorted lattice⁴², matching with model-based EXAFS fitting (Supplementary Fig. 25 and Supplementary Table 3). Ni and $Fe K$ -edge FT-EXAFS spectra were fitted to the NiFe oxyhydroxide structure with comparable $Ni-O$ and $Fe-O$ average bond length (Supplementary Figs. 26, 27 and Supplementary Table 4)^{44,43}. Formation energy calculations indicated that in-layer substitution with octahedral WO_6 configuration was energetically favored over forming WO_4 tetrahedron anchored on the NiFeOOH basal plane (Supplementary Fig. 28).

The second derivatives of the $W L_3$ -edge XANES spectra supported that distorted octahedral WO_6 configuration, similar to WO_3 was observed in W_1 -NiFeOOH, rather than the tetrahedral coordination of the original Na_2WO_4 precursor (Fig. 2f), suggesting a local structural transformation and abrupt substitution into oxyhydroxide lattice during the simultaneous electrodeposition process. Wavelet transform (WT) analysis of EXAFS clarified that the local atomic

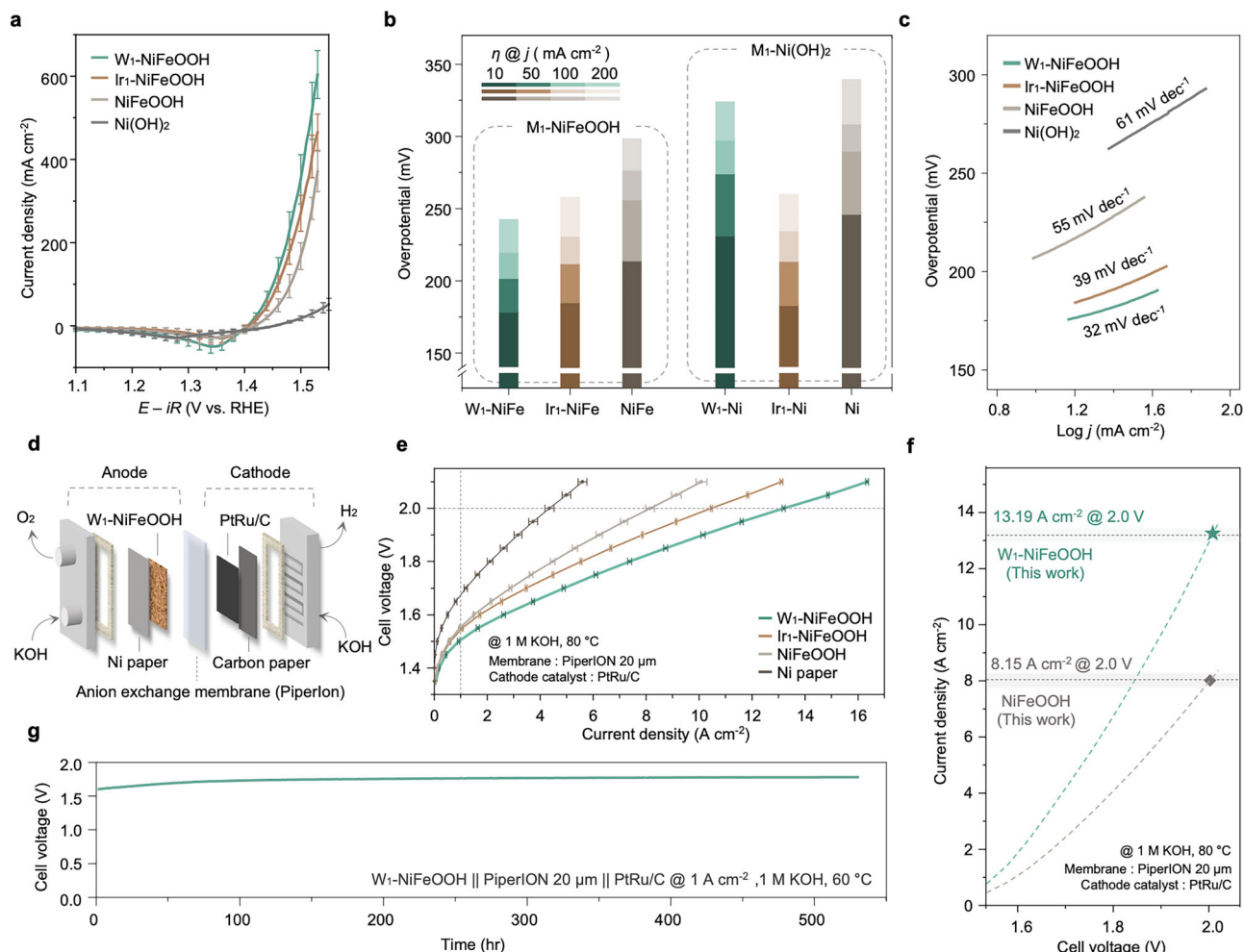


Fig. 3 | Electrocatalytic OER performance. **a** OER polarization curves in 1 M KOH (pH = 13.7) electrolyte using a half-cell three-electrode setup with iR compensation. E is the measured potential versus RHE, and iR is the ohmic drop. The scan rate was set to 5 mV s^{-1} , and the electrolyte temperature was 25°C . Error bars represent the standard deviation from three independent measurements ($n = 3$). **b** Comparison of overpotentials at specific current densities. **c** Tafel plots derived from the polarization curves. **d** Schematic illustration of the AEMWE cell configuration. **e**

Polarization curves of the AEMWE electrolyzer in 1 M KOH electrolyte at 80°C without iR compensation. The scan rate was set to 10 mV s^{-1} , and the electrolyte flow rate was 10 ml min^{-1} for both anolyte and catholyte. Error bars represent the standard deviation from ten consecutive measurements ($n = 10$). **f** Comparison of AEMWE performance. **g** Chronopotentiometric stability test of the AEMWE electrolyzer measured at 1 A cm^{-2} and 60°C . Source data for Fig. 3a–c, e–g are provided as a Source data file.

structure around W atoms was not related to the WO_3 reference peak present in local maximum derived from W–O–W second shell scattering (Fig. 2g). In contrast, Ir atoms in $\text{Ir}_1\text{-NiFeOOH}$ showed valence states between +0 and +3, as indicated from Ir L_3 -edge XANES spectra (Supplementary Fig. 29). The structural configuration of anchored Ir-O_6 on the NiFeOOH basal plane was proposed based on Ir L_3 -edge FT-EXAFS results (Supplementary Fig. 30).

Electrocatalytic alkaline oxygen evolution reaction performance

Catalytic measurements were carried out in a three-electrode half-cell setup using 1 M KOH as the electrolyte to enable systematic assessment of electrocatalytic OER performance. To accurately evaluate the intrinsic activity, the cathodic scan of the polarization curve was employed to avoid interference from Ni redox transitions, as the $\text{Ni}^{3+} \rightarrow \text{Ni}^{2+}$ reduction peak appears at a lower potential and does not overlap with the OER onset region³⁴ (Supplementary Fig. 31). In contrast, the anodic scan presents a $\text{Ni}^{2+} \rightarrow \text{Ni}^{3+}$ oxidation peak in close proximity to the OER onset, complicating deconvolution of the electrocatalytic response. Accordingly, OER-associated currents and overpotentials were obtained from the cathodic polarization curves

(Fig. 3a). Among the tested catalysts, electrodeposited $\text{W}_1\text{-NiFeOOH}$ achieved the onset overpotential of 178 mV at 10 mA cm^{-2} , which was lower than the values obtained for $\text{Ir}_1\text{-NiFeOOH}$ (183 mV), NiFeOOH (214 mV), and a commercial IrO_2/C powder-based electrode (Supplementary Fig. 32).

Comparative analysis of overpotentials between $\text{M}_1\text{-Ni(OH)}_2$ and $\text{M}_1\text{-NiFeOOH}$ revealed significant discrepancies depending on the incorporated single-atom metal (Fig. 3b). The inclusion of Ir single atoms led to a substantial and consistent magnitude of improvement in catalytic performance, regardless of whether the support (oxy)hydroxide was Ni(OH)_2 or NiFeOOH . This suggests that Ir atoms themselves might act as dominant active sites, overwhelming Ni-related original active motifs in the metal (oxy)hydroxide. In contrast, while $\text{W}_1\text{-Ni(OH)}_2$ showed only a slight improvement over Ni(OH)_2 and remained inferior to pristine NiFeOOH , $\text{W}_1\text{-NiFeOOH}$ demonstrated enhanced OER performance, surpassing both $\text{Ir}_1\text{-Ni(OH)}_2$ and $\text{Ir}_1\text{-NiFeOOH}$ across all current densities. W atoms might further synergistically reinforce the active Ni–O–Fe sites in NiFeOOH edge plane, whereas the less active Ni–O–Ni sites in Ni(OH)_2 offered limited potential for improvement. A similar trend was observed in Co-based (oxy)hydroxides, where Mo single-atom substitution significantly

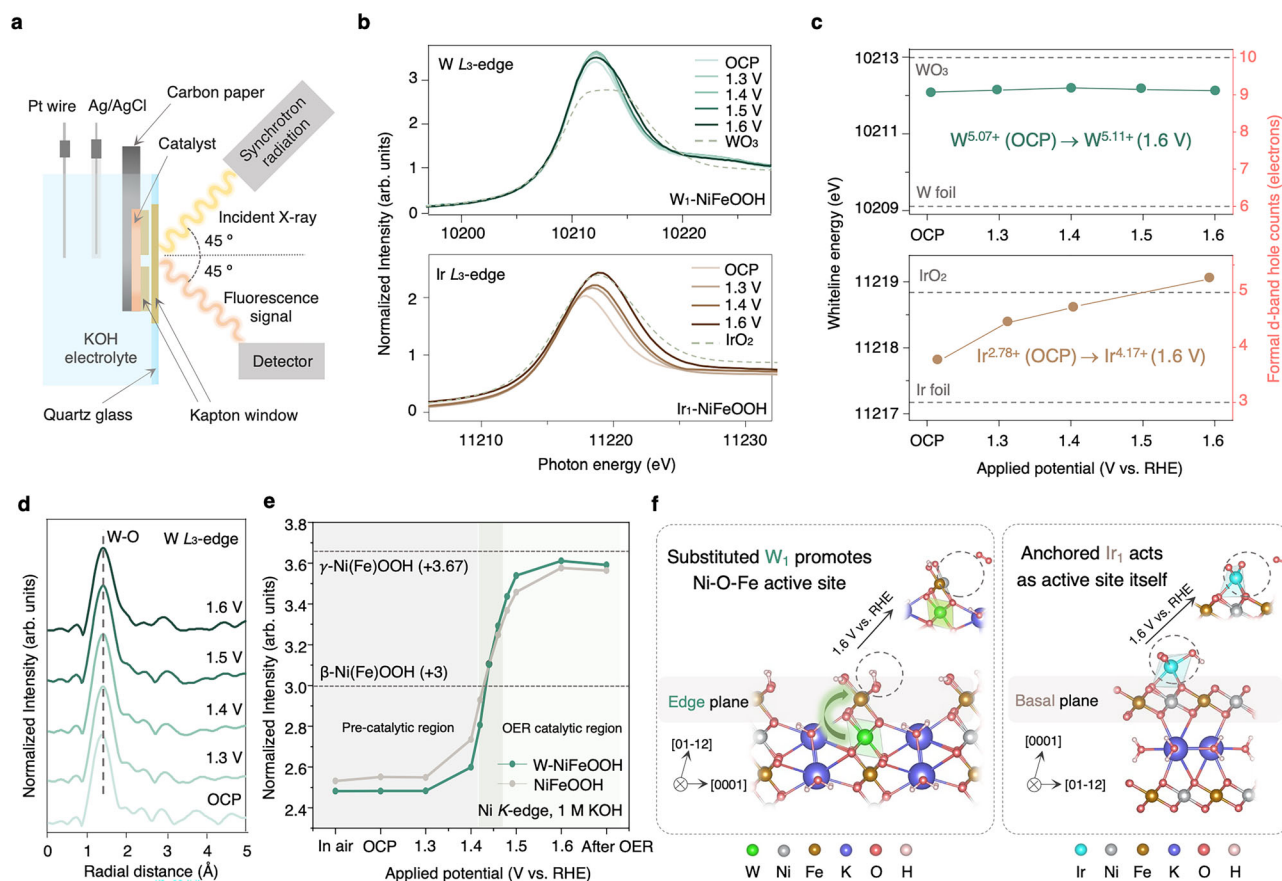


Fig. 4 | Revealing mechanism with in situ XAS spectroscopy. **a** Schematic illustration of in situ XAS measurement setup in fluorescence mode. **b** Normalized in situ W L_3 -edge and Ir L_3 -edge XANES spectra under various applied potentials in 0.1 M KOH using carbon paper as a substrate. **c** Valence state estimation under the relationships between d -band hole counts and the whiteline peak position of L_3 -edge in situ XANES spectra. **d** In situ W L_3 -edge FT-EXAFS with unshifted first-shell

W–O bonds related peaks. **e** Changes in the Ni oxidation states recorded from in situ Ni K -edge absorption edge shifts under 1 M KOH using carbon paper as a substrate. **f** Schematic illustration of proposed roles of W and Ir atoms in improving catalytic performance of W_1 -NiFeOOH and Ir_1 -NiFeOOH, respectively. Source data for Fig. 4b–e are provided as a Source data file.

enhanced OER activity only when incorporated into CoFeOOH host. (Supplementary Fig. 33).

Tafel slope analysis demonstrated enhanced OER kinetics for W_1 -NiFeOOH (Fig. 3c), and electrochemically active surface area (ECSA) measurements confirmed enhanced charge transfer with increased active site availability (Supplementary Fig. 34). Turnover frequency (TOF) was calculated as the minimum bulk limit (Supplementary Fig. 35) with near unity faradaic efficiency (FE) towards OER (Supplementary Fig. 36). Chronoamperometric stability tests demonstrated that W_1 -NiFeOOH maintained the initial current density for over 200 h (Supplementary Fig. 37). W_1 -NiFeOOH exhibited competitive electrocatalytic OER performance in the half-cell test as the state-of-the-art non-noble OER catalysts (Supplementary Fig. 38 and Supplementary Table 5).

To scale up towards ampere-level single-cell alkaline exchange membrane water electrolysis (AEMWE), we optimized the electrode-membrane interface and systematic membrane electrode assembly (MEA) conditions to minimize the internal cell resistance (Fig. 3d; Supplementary Figs. 39 and 40). AEMWE performance test at 80 °C in 1 M KOH demonstrated that W_1 -NiFeOOH requires only 1.56 V to reach 1 A cm⁻² and delivers 13.1 A cm⁻² at an applied cell voltage of 2.0 V, surpassing both NiFeOOH and Ir_1 -NiFeOOH counterparts (Fig. 3e). A similar trend was observed at 60 °C (Supplementary Fig. 41). W_1 -NiFeOOH demonstrated competitive catalytic performance and could be established as an efficient non-noble metal anode catalyst, exhibiting high activity among reported AEMWE systems employing

commercially available membranes (Fig. 3f and Supplementary Table 6). The long-term durability of the W_1 -NiFeOOH in AEMWE was further validated in 1 M KOH at 60 °C, achieving stable operation for over 500 h at 1 A cm⁻² (Fig. 3g), marking a significant step towards scalable green hydrogen production.

Unraveling OER mechanisms with in situ XAS and Raman spectroscopy

To investigate how atomically dispersed W or Ir atoms enhance the OER catalytic activity of metal oxyhydroxides, we utilized in situ XAS and Raman spectroscopy. These techniques enabled us to monitor real-time changes in the electronic states and local structural environments of the catalysts under operational conditions, providing insights into their dynamic behavior during electrochemical reactions⁴⁴. In situ XAS measurements were conducted in fluorescence mode using a three-electrode setup (Fig. 4a). The white-line peak positions were defined as the energies at which the second derivative of the XANES spectra equals zero, and then performed a linear fit with reference standard values to quantify the oxidation states of Ir and W.

For W_1 -NiFeOOH, normalized in situ W L_3 -edge XANES spectra displayed consistent whiteline positions and intensities under varying applied potentials from open circuit potential (OCP) to 1.6 V, indicating that the W valence state and electronic configuration remained unchanged during OER (Fig. 4b). In additions, the linear relationship between whiteline energy positions and d -band hole counts^{40,45}, interpolated from standard references, allowed us to estimate the

valence state of the metal element. Average W valence states of +5.07 at OCP and +5.11 at 1.6 V were estimated with a negligible shift (Fig. 4c). The steady oxidation state under applied potentials across the OER region aligned with the post-OER sample without agglomeration into metallic nanoparticles (Supplementary Fig. 42). The first-shell W–O bonding length remained unchanged, as confirmed by in situ W L_3 -edge FT-EXAFS spectra (Fig. 4d). The observed constancy in valence state and local structural environment suggest that atomically dispersed W atoms are not direct active sites in the electrocatalytic process (Supplementary Fig. 43).

In contrast, Ir L_3 -edge XANES spectra showed a significant increase in both whiteline peak intensity and position with increasing potential, indicating a higher oxidation state and more vacant states in the d -orbital of the Ir atoms under OER conditions (Fig. 4b). At OCP, the whiteline energy position implied a valence state of +2.17, interpolated from Ir foil ($5d^7$) and IrO₂ ($5d^5$) reference. As the applied voltage increased to 1.6 V, a dramatic positive shift in the whiteline energy suggests that isolated Ir atoms were oxidized to a valence state of +4.17, surpassing that of IrO₂ (Fig. 4c). A negative shift in Ir–O scattering path peak observed in the in situ FT-EXAFS spectra from OCP to 1.6 V implied an Ir–O bond contraction, reflecting structural changes from Ir–OH coordination to partially deprotonated O-terminated states with shortened Ir–O bonds and increased valence state (Supplementary Fig. 44)⁴⁶. These reaction intermediate-related dynamic changes in the local structural and chemical state verified that Ir atoms serve as direct active sites in the OER process.

To probe the oxidation behavior and structural evolution of Ni centers under OER conditions, in situ Ni K -edge XANES measurements were performed on W₁–NiFeOOH and NiFeOOH in 1 M KOH (Supplementary Fig. 45)⁴⁷. W₁–NiFeOOH exhibited a pronounced anodic shift in the Ni K -edge absorption edge beginning near 1.40 V vs. RHE, with significantly larger edge shifts than NiFeOOH at all potentials (Fig. 4e). This indicates not only an earlier onset of Ni oxidation but also a greater extent of oxidation, with the estimated average Ni valence reaching approximately +3.6, consistent with the formation of the catalytically active γ -NiOOH phase. In contrast, NiFeOOH required higher potentials to achieve comparable oxidation shifts, suggesting a higher energy barrier for γ -phase activation. The presence of W atoms thus facilitated both the onset and extent of Ni oxidation, promoting structural transformation into the active γ -NiOOH phase with a theoretical Ni oxidation state approaching +3.75. This bulk oxidation behavior was corroborated by ex-situ Ni $2p$ XPS measurements, which revealed a positive shift in binding energy after OER, indicating an increased surface Ni oxidation state (Supplementary Fig. 46). Corresponding in situ EXAFS analysis further revealed distinct structural reorganization in W₁–NiFeOOH, with shifts in Ni–O–M radial distances and coordination patterns indicative of α -to- γ phase transformation (Supplementary Fig. 47). These spectroscopic results confirm that W incorporation promotes the formation of the catalytically active phase by enabling more efficient and extensive Ni oxidation under OER-relevant potentials. The observed distinct roles of W and Ir single atoms lead to the different mechanisms through which these elements contribute to the enhanced catalytic performance of W₁–NiFeOOH and Ir₁–NiFeOOH compared to NiFeOOH, respectively (Fig. 4f).

To complement the in situ XAS results, we performed in situ Raman spectroscopy to investigate the phase transformation of metal oxyhydroxides during OER conditions. The in situ Raman spectra were collected in 1 M KOH under various applied voltages using a three-electrode setup (Fig. 5a). For W₁–NiFeOOH, the spectra revealed a phase transition from the as-prepared α -phase to the OER active γ -phase at 1.43 V (Fig. 5b), evidenced by the emergence of characteristic Raman peaks at 474 and 553 cm^{−1}, corresponding to the E_g bending and A_{1g} stretching vibration modes of Ni(III)–O in γ -NiOOH, respectively⁴⁸. As the applied potential increases, the significant peak shift from the initial peaks at 455 and 525 cm^{−1}, associated with Ni(II)–OH and Ni(II)–O

vibration modes in the α -phase Ni(OH)₂, indicated a dynamic structural transformation towards γ -phase^{20,37,38,43}. Notably, characteristic peaks related to γ -phase began to appear at lower potentials for W₁–NiFeOOH (1.43 V) compared to both Ir₁–NiFeOOH and NiFeOOH (1.45 V) (Fig. 5c and Supplementary Fig. 48). This implies that W atoms might contribute to the energetically favored transformation to the catalytically active γ -phase, consistent with the in situ Ni K -edge XANES results.

The extent of structural transformation towards the active γ -phase oxyhydroxide phase was determined by the intensity ratios of the characteristic Raman peaks⁴⁹. A notable transition in the peak intensity ratio was observed for W₁–NiFeOOH at lower potentials compared to Ir₁–NiFeOOH and NiFeOOH (Fig. 5d), indicating a more pronounced α -to- γ -phase transition, accompanied by the stabilization of surface Ni–O bonds in the presence of higher Ni oxidation states. The intercalation of K⁺ ions into the layered (oxy)hydroxide structure stabilizes the transformation from the inactive α -phase to the catalytically active γ -phase with a high Ni oxidation state under applied potentials (Fig. 5e). Consequently, in situ XAS and Raman spectroscopy measurements revealed that W promotes the original active site of NiFeOOH by facilitating the transitions in Ni oxidation state and the active γ -phase transformation at lower overpotentials, while Ir acts directly as an active site, leading to significantly enhanced OER catalytic performance.

Mechanistic insights from DFT calculations

To provide computational insights into the catalytic mechanisms by which atomically dispersed W and Ir enhance the intrinsic OER activity, we performed DFT calculations that provide insights into the electronic structure modifications and the modulated adsorptions of reaction intermediates.

For W₁–NiFeOOH, the proposed dominant OER pathways were based on the Mars-van-Krevelen mechanism (Supplementary Fig. 49), while Ir₁–NiFeOOH proceeded via an adsorbate evolution mechanism involving surface-bound oxygen intermediates (Fig. 6a)^{34,50}. The spin state and magnetic moment analysis indicated the oxidation states of the active sites (Fig. 6b) and implied that the substitution of subsurface W atoms into the NiFeOOH lattice modified the electronic structure, promoting the formation of highly active Ni–O–Fe bridge sites at the edge plane. These configurations stabilize the deprotonation of the adsorbed OH, thereby lowering the energy barrier for the *OH to *O transition and facilitating the OER process.

The Gibbs free energy diagrams for OER on edge active sites (Fig. 6c) showed that the free energy barrier for the potential-determining step on W₁–NiFeOOH, specifically the *OH to *O transition at Ni–O–Fe bridge sites with subsurface W on the edge plane, was markedly lower compared to that of pristine NiFeOOH and Ir₁–NiFeOOH. The exposed W–O–Ni/Fe motifs on the edge plane or W substituted on the basal plane exhibited a higher barrier than the original Ni–O–Fe active sites (Fig. 6d), suggesting that W atoms themselves are not the primary active sites but rather act as promoters.

This result is consistent with experimental observations from in situ Raman spectra after 1 M TMAOH treatment (Supplementary Fig. 50) and pH-dependent OER kinetics (Supplementary Fig. 51), which support active lattice oxygen participation. Our results corroborate that W acts as a promoter in W₁–NiFeOOH, facilitating a Mars-van-Krevelen-type lattice oxygen mechanism that preferentially occurs at Ni–O–Fe bridge sites in the edge plane by stabilizing the O* intermediate and lowering the overall reaction barrier.

In contrast, anchored Ir–O motifs on the basal plane in Ir₁–NiFeOOH served as the active sites with the lowest energy barrier, which is proposed to contribute to the enhanced OER activity. To better capture the alkaline electrochemical environment, we extended our DFT analysis to include multiple hydroxylated Ir anchoring motifs. The Ir₁–3O configuration exhibited the lowest overpotential of 0.310 V,

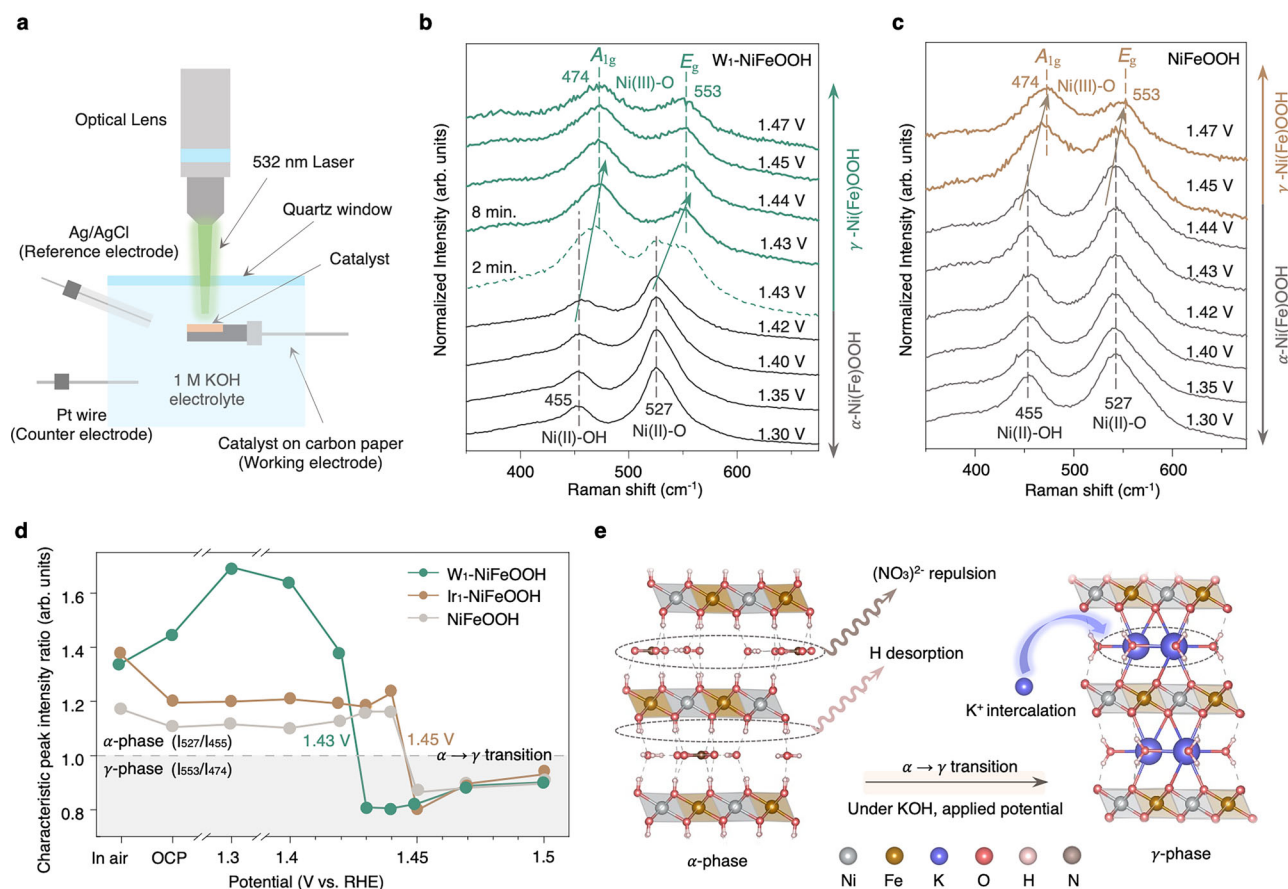


Fig. 5 | Structural phase transformation under in situ Raman spectroscopy. **a** Schematic illustration of the in situ Raman measurement setup. Measurements were done under 1 M KOH using carbon paper as a substrate. **b**, **c** Raman spectra recorded under increasing applied potentials from 1.30 to 1.47 V vs. RHE without *iR* correction was recorded with **b** W_1 -NiFeOOH and **c** NiFeOOH. E_g and A_{1g} denote the Ni-O bending and stretching vibrational modes, respectively. The phase transformation of NiFe (oxy)hydroxide was demonstrated via blue-shifted characteristic

peaks. **d** Raman characteristic peak intensity ratios (I_{527}/I_{455}) and (I_{553}/I_{474}) under applied potentials, where I_{553} and I_{474} represent the peak intensities at 553 and 474 cm^{-1} , while I_{527} and I_{455} represent the peak intensities at 527 and 455 cm^{-1} , which indicates the transition between α and γ phase. **e** Schematic illustrating γ -phase transition of NiFe oxyhydroxide accompanied by K^+ ion intercalation. Source data for Fig. 5b–d are provided as a Source data file.

while the Ir_1 -3OH configuration showed a notably higher value of 0.485 V, both with the O* to *OOH step being potential-determining (Supplementary Fig. 52). The broader range of predicted overpotentials from these additional geometries helps reconcile the apparent discrepancy between DFT calculations and the experimentally observed enhanced activity of W_1 -NiFeOOH relative to Ir_1 -NiFeOOH.

Furthermore, the projected density of states analysis supported that W substitution modulates the electronic structure of the pristine NiFeOOH (Fig. 6d). Specifically, the incorporation of W led to an upward shift in the Ni 3d band center and increases the overlap with O 2p orbitals, thereby stabilizing the formation of surface metal-oxygen adsorbate bonds. These shifts modulated the adsorption energies of oxygen intermediates, facilitating the crucial *OH to *O transition with hybridization between Ni 3d and O 2p states and improving the overall catalytic performance of the metal (oxy)hydroxide (Fig. 6e)²⁵.

Discussion

We efficiently identified W_1 -NiFeOOH as the most promising noble metal-free OER catalyst candidate by accelerated ML-guided catalyst screening and experimental validation of its electrocatalytic performance with unambiguous catalytic mechanisms. Accelerated adsorption energy evaluations using the pretrained EquiformerV2 model pinpointed W_1 -NiFeOOH as a promising noble metal-free candidate. The optimized electrodeposition method enabled the scalable and

rapid growth of W_1 -NiFeOOH nanosheets with a consistent high-loading of atomically dispersed W atoms and validated with competitive OER performance among non-noble metal catalysts. W_1 -NiFeOOH achieved a high current density of 13.1 A cm^{-2} at 2.0 V (80 °C) in AEMWE using commercially available membranes, with robust operational stability exceeding 500 h. These results establish W_1 -NiFeOOH as a state-of-the-art non-noble metal anode catalyst for scalable green hydrogen production. The role of distinct active motifs on catalytic mechanisms was revealed through in situ XAS and Raman spectroscopy, supported by DFT calculations, that substitutional W atoms promote the original edge active sites of NiFe oxyhydroxide by facilitating deprotonation at the original NiFeOOH edge active sites and the active γ -phase transformation at lower potentials, unlike the direct participation of anchored Ir single atoms in Ir_1 -NiFeOOH. This study bridges computational predictions with experimental validation to accelerate the discovery of highly active non-precious metal catalysts and pinpoint the underlying mechanism, paving the way for designing competitive catalytic solutions for sustainable global energy conversion.

Methods

Chemicals and materials

Nickel (II) nitrate hexahydrate ($Ni(NO_3)_2 \cdot 6H_2O$), cobalt (II) nitrate hexahydrate ($Co(NO_3)_2 \cdot 6H_2O$), iron (II) sulfate heptahydrate ($FeSO_4 \cdot 7H_2O$), sodium tungstate dihydrate ($Na_2WO_4 \cdot 2H_2O$), sodium



(saturated KCl) was carried out at the scanning rate of 50 mV s^{-1} for 30 cycles. The deposited electrode was rinsed with deionized water and absolute ethanol three times and dried at 25°C overnight. For the synthesis, $\text{M}_1\text{-NiFeOOH}$ ($\text{M}=\text{Ir}$, Ru , and Mo), $\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$ was replaced with $\text{IrCl}_3 \cdot x\text{H}_2\text{O}$ (0.1 mmol), $\text{RuCl}_3 \cdot x\text{H}_2\text{O}$ (0.1 mmol), or $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$ (0.2 mmol). The electrodeposition procedures were the same as these for the synthesis of $\text{W}_1\text{-NiFeOOH}$. For the synthesis of $\text{M}_1\text{-Ni(OH)}_2$, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ was excluded and $\text{Ni(NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (10.5 mmol) was dissolved into solution. Precursor for M_1 was removed to prepare pristine NiFeOOH or Ni(OH)_2 . Co-based counterparts were prepared by replacing $\text{Ni(NO}_3)_2 \cdot 6\text{H}_2\text{O}$ with $\text{Co(NO}_3)_2 \cdot 6\text{H}_2\text{O}$.

Material characterization

Field emission scanning electron microscope (MERLIN Compact, ZEISS) was used to identify the surface morphology of the sample with an accelerating voltage of 5 kV. Spherical aberration corrected monochromated scanning TEM (Themis Z 60–300, Thermofisher) operated at an accelerating voltage of 300 kV was used to obtain high-angle annular dark field-scanning (HAADF) scanning TEM images and energy dispersive X-ray mapping spectra. AFM (model) was used to determine the thickness of the nanosheet sonicated on the atomically smooth Si(111) wafer substrate. XPS analysis was performed using an ULVAC-PHI spectrometer (PHI5000 Versa Probe II) equipped with an Al $K\alpha$ (1486.6 eV) radiation source. The background of the XPS spectra was corrected with the Shirley method, and the peaks were deconvoluted and fitted using CasaXPS software. High-resolution XRD patterns were obtained using a (D8 Advance, Bruker) with Cu $K\alpha$ radiation. To improve the detection of weak diffraction signals from the (oxy) hydroxide phase and suppress the dominant peaks from the carbon paper substrate, data were collected in logarithmic intensity scale at a scan rate of 2° min^{-1} while continuously rotating the sample at 30 rpm. To prepare ICP-AES sample solution, electrocatalysts were electrodeposited on a carbon paper and dissolved in the 3:1 mixture of HCl and HNO_3 and heated to 80°C for 1 h. The dissolved solution was cooled down to 25°C and diluted with deionized water.

Electrochemical measurements

The electrocatalytic OER measurements were carried out using a potentiostat (Ivium-n-Stat, Ivium Technologies) in the three-electrode cell with catalyst-electrodeposited Ni foam, Ag/AgCl (saturated with KCl, ALS Japan), and Pt plate as the working, reference, and counter electrode, respectively. The 1 M KOH electrolyte (pH 13.7) was stored in a sealed polypropylene bottle to prevent CO_2 absorption and contamination. For all electrochemical measurements, fresh electrolyte was used immediately after dispensing. All of the recorded potentials in 1 M KOH electrolyte were converted from Ag/AgCl scale into reversible hydrogen electrode (RHE) scale by the following Nernst equation:

$$E_{\text{RHE}} = E_{\text{Ag/AgCl}} + E_{\text{Ag/AgCl}}^0 + 0.059 \times \text{pH} \quad (1)$$

where $E_{\text{Ag/AgCl}}$ is the measured potential (V) versus the Ag/AgCl electrode, and $E_{\text{Ag/AgCl}}^0$ is 0.198 V at 25°C . Water oxidation was measured recording a linear sweep voltammetry (LSV) from 1.1 to 1.8 V versus RHE with a scan rate of 5 mV s^{-1} , and successive cathodic linear sweeping from 1.8 to 1.1 V versus RHE was recorded three times, after CV activation of 20 cycles at 20 mV s^{-1} . System resistances were determined from the high-frequency intercept of the Nyquist plots on the real axis, obtained via electrochemical impedance spectroscopy in a frequency range from 100 kHz to 0.1 Hz. The extracted resistance values are provided in the Source data file. For the three-electrode half-cell measurements, the potentials were iR -corrected for 95% of the measured cell resistance. The ECSA was estimated from the double-layer capacitance (C_{dl}) determined by CV measurements in the non-faradaic region. The capacitive current density at the center of the

potential window was plotted against the scan rates, where the linear slope represented C_{dl} . ECSA was calculated by dividing C_{dl} with a specific capacitance (C_s) of $40 \mu\text{F cm}^{-2}$ for alkaline conditions. The FE of OER was evaluated at a constant potential of 1.5 V vs. RHE without iR compensation and derived from a single measurement using the following equation:

$$\text{FE}(\%) = 100 \times (4 \times F \times n_{\text{O}_2}) / Q \quad (2)$$

where 4 is the number of electrons transferred per O_2 molecule, F is the Faraday constant ($96,485 \text{ C mol}^{-1}$), n_{O_2} is the number of moles of measured O_2 , and Q is the total charge passed during the reaction. TOF values were derived from a single measurement using the following equation:

$$\text{TOF} = (j \times A) / (4 \times F \times n) \quad (3)$$

where j is the current density at 1.5 V vs. RHE with iR correction, A is the geometric area of the electrode, F is the Faraday constant ($96,485 \text{ C mol}^{-1}$), and n is the total number of moles of metal atoms deposited on the electrode, as determined by ICP-AES. To provide a lower limit of TOF, it was assumed that every metal atom identified via ICP-AES participated as a catalytic active site.

AEMWE cell was composed with Ni-coated stainless-steel plates with serpentine channel, and the MEA was prepared stacking the anode, anion exchange membrane (AEM), and cathode with the applied pressure of 55 N m^{-1} . Electrodeposited $\text{W}_1\text{-NiFeOOH}$ and NiFeOOH on nickel paper were utilized as anodes. For the cathode, a catalyst ink was prepared by dispersing PtRu/C in the mixture of isopropanol solution and PiperION ionomer solution, followed by ultrasonication to ensure a homogeneous suspension. This ink was airbrushed onto a PTL-coated carbon paper with a geometric area of 4.4 cm^2 , and the catalyst loading was controlled to reach 1.0 mg cm^{-2} , which was verified by measuring the mass difference of the substrate before and after the airbrushing and drying process. The commercial $20 \mu\text{m}$ -thick AEM (PiperION, Versogen) was soaked in 1 M KOH for at least 24 h, followed by rigorous rinsing with distilled water. Both the cathode and anode were cut to the size of 4.4 cm^2 . The AEMWE single cell was operated at temperatures of 60 and 80°C , and heated 1 M KOH electrolyte was circulated by at a flow rate of 10 mL min^{-1} . The polarization curves of the AEMWE single cell were obtained using a potentiostat (HCP-803, BioLogic) with LSV at a scan rate of 50 mV s^{-1} in the cell voltage range of 1.3–2.1 V. The system resistance of the AEMWE single cell was determined from EIS performed at a current density of 0.5, 1, 2.5, and 5 A cm^{-2} . The raw EIS data for each current density was provided in the Source data file. The polarization curves for the AEMWE were reported without iR -correction.

XAFS measurements

X-ray absorption fine structure (XAFS) measurements were performed at the nanoprobe XAFS beamline (BL8C) of the Pohang light source (PLS-II) in the 3.0 GeV storage ring with a ring current of 300 mA. The incident X-ray beam was monochromatized by a Si(111) double crystal, and the beam intensity was reduced by 30% to eliminate higher-order harmonics. The photon energy was calibrated before the measurement using a reference foil. XAFS spectra were collected in both transmission and fluorescence modes by a solid-state detector. The acquired XAS data were processed using Demeter software integrated with the IFEFFIT software package. In situ XAS measurements were performed in custom-made electrochemical cell with three electrode setup and KOH electrolyte. The Kapton tape window was employed for incident X-ray beam penetration.

Raman spectroscopy measurements

The Raman measurements were carried out on an optical microscope combined with a Raman spectrometer (Horiba Jobin Yvon, LabRAM ARAMIS) with a 532 nm excitation laser and calibrated with Si reference before all measurements. In situ Raman measurements were employed in a custom-made electrochemical cell with the three-electrode setup.

Catalyst material cost evaluation

We considered only the raw metallic element cost to evaluate the catalyst raw material cost. For M_1 -NiFeOOH, we assumed that metal content loadings of 5 mg cm⁻² and 1 atomic % of M atoms were included in the NiFe oxyhydroxide supports with atomic ratio of Ni and Fe as 3:1. For IrO₂, we assume the metallic iridium content as 40 wt%, matching with the commercial IrO₂/C catalyst. The raw elemental metal costs were obtained as follows, Ni (-15.97 USD per kilogram, Shanghai Metals Market, 20 August 2024), Fe (-0.61 USD per kilogram, Shanghai Metals Market, 20 August 2024), W (-45.07 USD per kilogram, Shanghai Metals Market, 20 August 2024), Ir (-162,577 USD per kilogram, Shanghai Metals Market, 20 August 2024), Rh (-170,705 USD per kilogram, Shanghai Metals Market, 20 August 2024), and Ta (-345.76 USD per kilogram, Shanghai Metals Market, 20 August 2024).

Machine learning processes

The ML framework was implemented using the EquiformerV2 model²⁸, pretrained on the OC22 dataset. The model architecture consists of 18 layers with 8 attention heads, employing 128 sphere channels and 128 hidden channels for the feed-forward networks. The maximum degree of representations was set to 6 and the maximum order to 2. Spherical activations were performed using 128 sphere samples with a grid resolution of 18, and attention re-normalization and separable layer normalization were applied. Slab models with various adsorption configurations were constructed with the ASE package⁵¹. High-throughput inference was performed using an NVIDIA A6000 GPU environment, applying the pretrained EquiformerV2 model to rapidly evaluate the total energies and atomic forces of the model systems iteratively towards the equilibrium geometry with the BFGS algorithm, with residual maximum force on each atom were smaller than 0.03 eV Å⁻¹. Adsorption energies-based reaction step barriers were calculated using:

$$dE_{\text{DFT}}(\text{OH} \rightarrow \text{O}) = E_{\text{DFT}}(^*\text{O}) - E_{\text{DFT}}(^*\text{OH}) + 0.5E_{\text{DFT}}(\text{H}_2) \quad (4)$$

$$dE_{\text{DFT}}(\text{O} \rightarrow \text{OOH}) = E_{\text{DFT}}(^*\text{OOH}) - E_{\text{DFT}}(^*\text{O}) - (E_{\text{DFT}}(\text{H}_2\text{O}) - 0.5E_{\text{DFT}}(\text{H}_2)) \quad (5)$$

$$dE_{\text{DFT}}(\text{OOH} \rightarrow \text{V}) = 4.92 - E_{\text{DFT}}(^*\text{OOH}) + E_{\text{DFT}}(^*\text{V}) + (2E_{\text{DFT}}(\text{H}_2\text{O}) - 1.5E_{\text{DFT}}(\text{H}_2)) \quad (6)$$

$$dE_{\text{DFT}}(\text{V} \rightarrow \text{OH}) = E_{\text{DFT}}(^*\text{OH}) - E_{\text{DFT}}(^*\text{V}) - (E_{\text{DFT}}(\text{H}_2\text{O}) - 0.5E_{\text{DFT}}(\text{H}_2)) \quad (7)$$

$$dE_{\text{ML}}(\text{OH} \rightarrow \text{O}) = E_{\text{ML}}(^*\text{O}) - E_{\text{ML}}(^*\text{OH}) + 0.5E_{\text{DFT}}(\text{H}_2) \quad (8)$$

$$dE_{\text{ML}}(\text{O} \rightarrow \text{OOH}) = E_{\text{ML}}(^*\text{OOH}) - E_{\text{ML}}(^*\text{O}) - (E_{\text{DFT}}(\text{H}_2\text{O}) - 0.5E_{\text{DFT}}(\text{H}_2)) \quad (9)$$

$$dE_{\text{ML}}(\text{OOH} \rightarrow \text{V}) = 4.92 - E_{\text{ML}}(^*\text{OOH}) + E_{\text{ML}}(^*\text{V}) + (2E_{\text{DFT}}(\text{H}_2\text{O}) - 1.5E_{\text{DFT}}(\text{H}_2)) \quad (10)$$

$$dE_{\text{ML}}(\text{V} \rightarrow \text{OH}) = E_{\text{ML}}(^*\text{OH}) - E_{\text{ML}}(^*\text{V}) - (E_{\text{DFT}}(\text{H}_2\text{O}) - 0.5E_{\text{DFT}}(\text{H}_2)), \quad (11)$$

where *X denotes the adsorbed intermediate ($X = \text{V}, \text{OH}, \text{O}$, or OOH). Fine-tuning of the pretrained model was conducted exclusively on the M_1 -NiNiOOH subset to assess transferability to binary compositions. Among the hyperparameter combinations tested, the selected configuration was trained on an NVIDIA A100 GPU using a batch size of 8 and a dropout rate of 0.1 for 12 epochs (Supplementary Table 1). The model was optimized using the AdamW algorithm with an initial learning rate of 0.0002, weight decay of 0.001, and a cosine annealing schedule with linear warm-up. Energy and force loss coefficients were set to 4 and 100, respectively, in accordance with OC22 training protocols. Gradient clipping (norm threshold = 50) and an exponential moving average decay of 0.999 were applied for training stability.

DFT calculations

The spin-polarized DFT calculations were performed with the projected augmented wave method⁵² with Perdew–Burke–Ernzerhof exchange correlation functional⁵³, as implemented in the Vienna ab initio simulation package⁵⁴. To correct for self-interaction errors of d -electrons, we employed the DFT + U method. The effective Hubbard U values (U_{eff}) were taken as 5.20 and 2.56 eV for Ni and Fe from the previous literatures^{10,34}, and 6.20 eV for W from the Materials Project database⁵⁵. A functional of DFT-D3 method of Grimme with zero-damping function was applied to consider van der Waals interactions⁵⁶. For a slab model, cutoff energy of 500 eV was used for the plane-wave expansion and Monkhorst-Pack ($1 \times 6 \times 4$) $\mathbf{k}$ -point grid was used for Brillouin zone integration $\mathbf{k}$ -point grids. The large vacuum layers were set at least 20 Å in the direction perpendicular to the surface to prevent the interference between two periodic unit cells with dipole correction. The bottom atomic monolayers of basal (0001) plane slab and edge (01–12) plane slab were fixed at their bulk positions, while the remaining atomic layers adsorbates were free to move in all directions until the total energy convergence and residual maximum force on each atom were smaller than 1×10^{-5} eV and 0.01 eV Å⁻¹, respectively. The equilibrium geometries were obtained was achieved for the electronic self-consistent field loop.

Data availability

The data supporting the findings of this study are available within the article and its Supplementary Information. Source data are provided with this paper.

Code availability

The model training configurations and pretrained checkpoints generated in this study are available on Zenodo at <https://doi.org/10.5281/zenodo.18205124>⁵⁷.

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Author contributions

J.K., S.H., P.S., Y.K., and H.W.J. conceived the project. J.K. performed catalyst screening, DFT calculations, synthesized and characterized the catalysts, carried out electrochemical measurements and in situ Raman measurements, and analyzed the overall experimental and

computational results. J.K. and I.S.K. performed in situ XAS measurements and analyzed the XAS results. J.K. and J.L. carried out MEA fabrication and AEMWE measurements. S.A.L. helped to optimize the electrodeposition method. S.P. carried out the AFM characterization. Y.J.K. helped to set up the computing environment. J.C., W.S.C., M.G.L., and K.C.K. provided advice on electrochemical measurements. H.W.J. supervised the project. J.K. and H.W.J. mainly wrote the manuscript. All authors discussed the results and commented on the manuscript at all stages.

Competing interests

The authors declare no competing interests.

Additional information

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